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Vision-enhanced multi-modal learning framework for non-destructive pavement damage detection

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ABSTRACT

Road infrastructure damage detection is crucial for maintaining transportation safety and efficiency. While current methods like machine learning and computer vision are effective, they have difficulty analyzing and describing complex pavement conditions. This paper presents DamageQwen, a vision-language model framework for non-destructive testing of pavement damage. The framework integrates YOLO v11x for initial damage detection with vision-language models (VLMs) to enhance detection accuracy and reduce reliance on large labeled datasets. By combining object detection capabilities with natural language understanding, DamageQwen enables more comprehensive damage assessment through visual question answering. The framework incorporates CLIP-based repeat defects elimination and supports few-shot learning with just 1–4 shots. Compared to generic VLMs, our approach significantly improves the accuracy of descriptions while greatly reducing the labelling effort. The framework's ability to process both visual and semantic information allows non-experts to effectively detect and assess pavement damage through simple prompts.

1. Introduction

Road infrastructure has a significant impact on a country's economic development and the efficiency of its transportation system. Well-maintained roads can significantly reduce the incidence of traffic accidents, thereby enhancing the safety of both drivers and pedestrians. Moreover, it can diminish vehicle wear and tear and fuel consumption, thereby lowering transportation costs for companies or individuals. However, disruptions to such critical transport infrastructure, whether due to aging, natural hazards, or other factors, can lead to severe socio-economic consequences. Therefore, enhancing the resilience of these transport networks through proactive maintenance regimes—informed by risk-based assessments and long-term performance monitoring—is of paramount importance [1]. Consequently, maintaining roads in optimal condition through regular inspections and repairs remains crucial.

Most current highway infrastructure inspections are performed manually or through inspection vehicles [2]. Inspectors can detect subtle damage based on experience and are able to adapt their inspection methods to the operating site environment. However, the periodic inspection tasks require a large number of human resources, which can significantly increase the inspection costs. Moreover, variability in damage assessment among different inspectors can result in significant errors in the final results. To address these shortcomings and improve the efficiency of infrastructure maintenance, automated inspection and damage detection methods offer significant advantages. These technologies, which can include Unmanned Aerial Vehicles (UAVs), machine learning (ML) analysis, and other advanced sensing techniques, provide unique capabilities for objective, consistent, and scalable assessment of critical infrastructure [3]. Among these automated solutions, multi-functional road damage inspection vehicles are ideally suited for large-scale road infrastructure inspection tasks. Inspection vehicles are usually equipped with advanced sensors and cameras, which can realize automated data collection and analysis tasks, and the judging standard is consistent, which can effectively reduce human errors. At the same time, inspection vehicles can simultaneously detect components such as pavement, subgrade and slope, thus providing more comprehensive information on road infrastructure conditions. However, the high cost of purchasing such vehicles is often prohibitive for financially constrained governments or management departments [4].

Therefore, engineers are beginning to investigate alternative inspection methods to improve efficiency and reduce inspection costs. Image processing is one of the most important techniques [5,6].

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Pavement damage detection technology based on image processing is mainly through the camera to collect pavement images, combined with image processing algorithms to realize the automatic identification and analysis of defects. Common techniques include edge detection [7], signal processing [8], and threshold segmentation [9]. The application of image processing technology enables the rapid processing and analysis of substantial quantities of pavement damage images, thereby reducing labour costs. However, this method is heavily dependent on manually configured features. It suffers from significant reductions in accuracy under varying lighting or weather conditions, a critical limitation that urgently needs to be addressed.

To minimize manual intervention, computer vision techniques are employed for pavement damage detection, enabling the automation of this process [10]. Vision models based on convolutional neural networks (CNN) are widely used for damage classification [11], damage detection [12], and damage segmentation [13]. CNN can automatically extract features of the damage and have better robustness than image processing techniques. In addition, trained neural networks apply a consistent evaluation methodology based on their learned parameters. This consistency can significantly reduce the subjective errors inherent in human detection, making the results more stable and reliable. However, training CNN requires large manually labeled datasets. Generally, the larger the dataset, the better the model's performance. Consequently, developing a highly accurate pavement damage detection model still requires significant labour costs. The model also requires retraining to recognize new types of damage. While computer vision-based object detection techniques [12] can effectively localize damage (outputting bounding boxes and classes), they typically stop there, failing to provide detailed natural language descriptions about damage severity, specific characteristics (like crack width variations), or repair recommendations, nor do they offer interactive capabilities. This research aims to move beyond this simple detection paradigm, leveraging vision-language models to provide richer, more interpretable, and more actionable damage assessment information.

With the development of large language models (LLMs) [14], the aforementioned problems can be solved. Vision language models (VLMs) [15], which are LLMs with vision detection capabilities, are well-suited for pavement damage detection tasks. VLMs combine vision and text processing capabilities to enhance the understanding of complex scenes with multi-modal data. This enables the VLMs to process not only image information but also textual descriptions of the object, thereby enhancing the model's robustness. The VLMs obtained from training in large-scale datasets exhibits a robust zero-shot learning capability, meaning it can classify or detect pavement damage images without additional training. To enhance the accuracy of the VLMs in pavement damage detection, it can be fine-tuned with a limited number of images. Therefore, VLMs are very suitable for damage with few samples. However, since the VLMs are still under development, the accuracy in object detection tasks has not yet reached the level of a specially optimized object detector (i.e. YOLO [16], ResNet [17]).

In summary, both the object detection models and VLMs exhibit distinct strengths and weaknesses in the task of pavement damage detection. To overcome these limitations, this paper proposes DamageQwen, a framework that strategically integrates the precision of state-of-the-art object detection with the contextual understanding and data efficiency of VLMs, enhanced by multimodal filtering techniques. The main contribution of this paper can be summarized as:

1. We propose DamageQwen, a VLM-based framework for pavement damage detection that enables damage assessment through Visual Question Answering.
2. With YOLO [16] and CLIP [18] combined to generate prompts for VLMs, we further improve the recognition accuracy of VLMs.
3. Simple user prompts, such as "Describe this image.", combined with prompts generated by the framework itself, lead to an accurate characterization of the damage. This enables non-experts to utilize the framework for detecting pavement damage.
4. The model proposed in this study requires only a small number of training samples (1-shot, 2-shot, and 4-shot), yet achieves significant improvements in accuracy metrics, thereby substantially reducing the workload associated with data labelling.

The rest sections of this thesis are organized as follows: Section 2 introduces the related work on computer vision and VLMs in pavement damage detection. Section 3 presents the VLMs framework proposed in this study. Section 4 discusses the related dataset and experimental configurations. Section 5 covers the results and discussion, and Section 6 concludes the paper.

2. Related work

In this section, we focus on pavement damage detection or segmentation algorithms based on computer vision and VLMs.

2.1. Pavement damage detection based on computer vision

With the development of computer vision, its application in structural health monitoring is becoming more and more widespread. The common methods are image classification, object detection and semantic segmentation.

Image classification can distinguish between images with and without defects. Wang et al. [11] were the first to apply convolutional neural networks to the classification of multi-class structural object damage, achieving excellent classification results. Authors in [19] integrated decision tree, logistic regression, and repeated incremental pruning to produce error reduction to achieve high-precision classification of longitudinal cracks, transverse cracks, and alligator cracks. Generally, pavement crack images are challenging to obtain. Therefore, [20] employed generative adversarial networks to extend the crack datasets and developed an enhanced VGG model that substantially improved the classification accuracy of crack images. With the development of classification models, the classification accuracy of defect images has even exceeded 99 %. However, classification results alone are insufficient in practical intelligent pavement inspection systems. To realize efficient automatic detection, the system not only needs to identify the type of damage, but also needs to accurately locate the location of the damage. Therefore, object detection algorithms that can simultaneously localize and classify defects hold greater research value and practical significance in the field of pavement damage detection.

Accurate damage detection and localization are crucial for enabling automated maintenance decisions and enhancing the efficiency of infrastructure management [21]. Du et al. [22] achieved 73.64 % accuracy based on the YOLO v3 model [23] in a self-constructed road damage dataset, as well as having a fast inference speed. Ren et al. [24] proposed a pavement damage detection model based on YOLO v5s while considering diagonal loss. Meanwhile, the study significantly improved the fusion between different features through a novel architectural design. As a result, the detection accuracy of the defect was significantly improved. Different backbone feature extraction networks were used by [25] for more complex multi-state pavement damage image detection, achieving an F1 score of 0.58 on the RDD2022 dataset [26], a result considered excellent when initial benchmarks for this dataset were being established. Zhang and Liu [27] embedded the deformable transformer into the YOLO v8 to improve the detection accuracy for irregular targets such as cracks. This algorithm achieves an F1 score of 0.601 on the RDD2022 dataset. By analyzing recent pavement damage detection algorithms, it is evident that these studies focus on designing more advanced or effective modules to further enhance the model's detection accuracy. Although object detection algorithms can effectively localize pavement defects, defects often exhibit irregular geometrical characteristics. Consequently, a simple rectangular detection box cannot

accurately capture the actual extent and boundary information of the defects [27]. This limitation may lead to the bias of disease area estimation, which affects the development of maintenance programs and cost assessment. This limitation may result in biased damage area estimation, thereby affecting the efficiency and accuracy of the pavement condition assessment process. In contrast, semantic segmentation technology can realize pixel-level damage recognition, which can accurately outline the damage, providing a more reliable quantitative basis for pavement maintenance decisions [28]. Therefore, applying semantic segmentation technology to pavement damage detection is crucial for enhancing accuracy and practicality.

The commonly used models in the field of pavement damage semantic segmentation include UNet [13], SegNet [29], HRNet [30], SegFormer [31] and so on. [32] proposed an improved damage segmentation model based on YOLO v7 to solve the missing feature problems. Zhang et al. [13] combined deformable convolution with the UNet to improve the model's ability to segment irregular shapes. To reduce the impact of noise elements like shadows on the segmentation results, [33] employed the discrete cosine transform to reduce shadow noise, thereby enhancing the robustness of the segmentation model. [34] combined the UNet model with the fully asymmetric loss, which well solved the imbalance problem between the object and the background, and thus provided accurate segmentation results for crack-like defects. The error of geometric features of cracks is kept within 16 %. It should be noted that this represents one of the most recent results concerning the detection of damage geometric features. More importantly, these vision-based methods typically do not generate natural language descriptions nor can they directly answer user queries about specific damages, limiting the communication and application of results to non-technical personnel. This motivates the exploration of new approaches that can fuse visual information with language understanding capabilities.

Assessing the severity of pavement damage involves a complex process of professional judgment. The same type of defect may present different severity levels depending on factors such as location, depth, and surrounding environment. Traditional computer vision methods often focus solely on the visual features of the defects, neglecting these crucial contextual factors. This single feature extraction approach is difficult to simulate the cognitive process of expert assessment, leading to a large discrepancy between the severity assessment results and the actual situation.

2.2. Pavement damage detection based on vision language models

In recent years, the rapid development of VLMs has provided new ideas to solve this problem [35]. Different from traditional methods, VLMs can understand both the visual content of images and related semantic information, aligning more closely with human cognition. By incorporating semantic information, the model not only recognizes the basic features of the damage but also understands its positional relationship within the pavement, potential impact, and other in-depth details. This multimodal understanding helps to construct a more comprehensive damage assessment system, which provides new possibilities for accurately estimating the severity of the damage.

Experimental results were used by Kumar, et al. [36] to demonstrate the potential of the GPT4 Turbo in detecting construction defects. [37] combined Segment Anything Model (SAM) [38] and Contrastive Language-Image Pre-training (CLIP) models [18] to realize the state of the art semantic segmentation results for building masonry. Pang, et al. [39] applied LLMs to the field of fault diagnosis of structures. Ferdousi, et al. [40] proposed a multimodal LLM to analyze visible and invisible visual defects in railroads and deploy it in consumer products. Yong, et al. [41] realized zero-shot and few-shot pavement damage classification based on CLIP model using prompt engineering approach. The authors confirmed from this study that multimodal information is more effective than unimodal information. Vision question answering models were also used by Kunlamai, et al. [42] for bridge damage detection. Users can provide images of bridges as input to the model, which then enables them to submit questions in natural language about the visual content. Subsequently, the model generates textual responses concerning bridge conditions.

Recent advancements in the field of industrial defect detection have increasingly utilized VLMs and LLMs techniques. These developments, which have predominantly emerged in 2024, align with the accelerated progress observed in both VLMs and LLMs over the last two years. Recent research has shown that VLMs predominantly excel in zero-shot or few-shot detection scenarios. Additionally, the multimodal understanding capabilities of these models facilitate the analysis of complex defect scenarios. However, VLMs are typically inferior to specialized visual models in terms of recognition accuracy for subtle defects. To enhance defect detection accuracy, integrating the high precision of specialized visual models with the multimodal comprehension of VLMs is recommended.

3. Proposed method

This section details the proposed DamageQwen framework, outlining its overall architecture, the defect pre-detection process using YOLO v11x, the CLIP-based module for eliminating redundant defects, the method for generating prompts.

3.1. Overall framework of DamageQwen

Current pavement damage detection faces a challenge: specialized object detection models, while precise in localization, struggle to describe complex damage situations and context, and rely on extensive labeled data; conversely, general VLMs offer descriptive power and some data efficiency but lack precision in locating specific small defects like cracks. Our proposed DamageQwen framework (Fig. 1) aims to bridge this gap through a multi-stage synergistic approach designed to leverage the strengths of different models. It begins with high-precision defect localization using an object detector, refines these detections via multimodal analysis, then uses this enhanced information to guide a VLM for detailed description generation, and finally fine-tunes the output with minimal examples. The overall workflow in Fig. 1 is as follows:

1. First, to ensure the precise localization capability that generic VLMs may lack, YOLO v11x identifies potential damage areas, providing initial detection and localization information.
2. Next, to address potential redundant boxes or misclassifications from the detector and enrich semantic understanding, the CLIP module filters and refines these initial detections.
3. The refined location and category information is then synthesized into a structured prompt, specifically designed to provide rich visually-grounded context to the VLMs.
4. Guided by these enhanced prompts, DamageQwen generates natural language descriptions of pavement damage that are more accurate and detailed than possible with only the raw image input.

3.2. Defects pre-detection

This subsection elaborates on the initial stages of damage detection within the DamageQwen framework, covering the use of the YOLO v11x object detector, the CLIP-based module for eliminating repeat defects, the generation of prompts for the Vision-Language Model, and the implementation of few-shot learning.

3.2.1. Object detector based on YOLO v11x

Recognizing that generic VLMs often struggle with precise localization of specific defect types like cracks, the first crucial step in our framework is accurate defect pre-detection. For this purpose, we employ

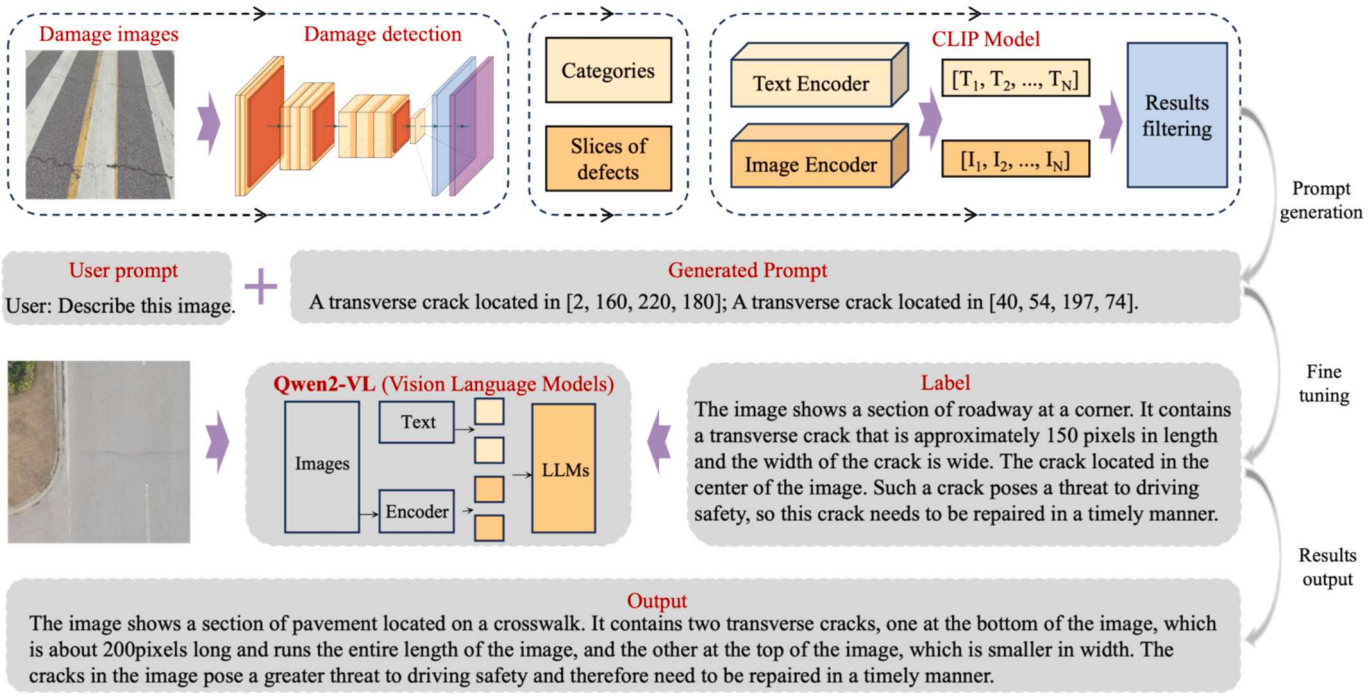


Fig. 1. Overall framework of DamageQwen.

YOLO v11x [43], one of the current SOTA object detectors, chosen for its high performance in identifying the location and initial category of defects within the pavement image. The processing of the YOLO v11x model is shown in Fig. 2. Defect images were first input into the YOLO v11x to obtain information about the locations and categories of the defects in the image. The resolution of the input image was limited to [224, 224]. The location information of the defects was represented by the coordinates of the upper-left and lower-right corners of the detection box in the image coordinate system, as illustrated in (1).

$$P_i = [x_{1i}, y_{1i}, x_{2i}, y_{2i}] \tag{1}$$

where P_i represents the location information of the i-th defect, $[x_{1i}, y_{1i}]$ denotes the upper-left coordinate of the i-th defect detection box, and $[x_{2i}, y_{2i}]$ denotes the lower-right coordinate of the i-th defect detection box.

After acquiring the primary detection results, we extracted the section of the input image within the detection boxes to get the defect images with their corresponding category information in the form of text.

3.2.2. *CLIP-based repeat defects eliminate module*

While YOLO v11x provides excellent initial detections, it can sometimes produce redundant bounding boxes or misclassify complex defects (as seen in Fig. 2), where a transverse crack might also be flagged as alligator). Standard Non-Maximum Suppression (NMS) struggles with duplicates across different categories. To overcome this limitation and further refine the inputs for the VLMs, we introduce a Repeat Defects Elimination Module (RDEM) based on the CLIP model [18]. CLIP is employed here specifically for its ability to leverage its multimodal understanding (connecting the semantics of the defect's image patch and its textual label) to filter redundancies that NMS might miss. Its strong zero-shot capability also aligns with our overall goal of minimizing annotation requirements. The configuration of this RDEM is depicted in Fig. 3.

The CLIP model is a multimodal model developed by OpenAI that integrates images and text features. It encodes images and text into the same feature space through contrastive learning, thereby establishing semantic associations between these different modalities. And it was trained using more than 400 million text-image pairs obtained from the Internet, so it possesses strong zero-shot capability, allowing it to be applied to the field of pavement defects detection without requiring fine-tuning. The main application scenarios of CLIP are currently in the field of zero-shot image classification, Image Retrieval, and Image-Text Matching. In this study, the multimodal and zero-shot learning capabilities of the CLIP model are primarily employed to eliminate redundant pavement damage detection boxes.

For an input image, YOLO v11x provided both the textual category information of the defect and a cropped image segment of the defect. These outputs were subsequently fed into the text encoder and image encoder of the CLIP model, respectively. For text input, the category

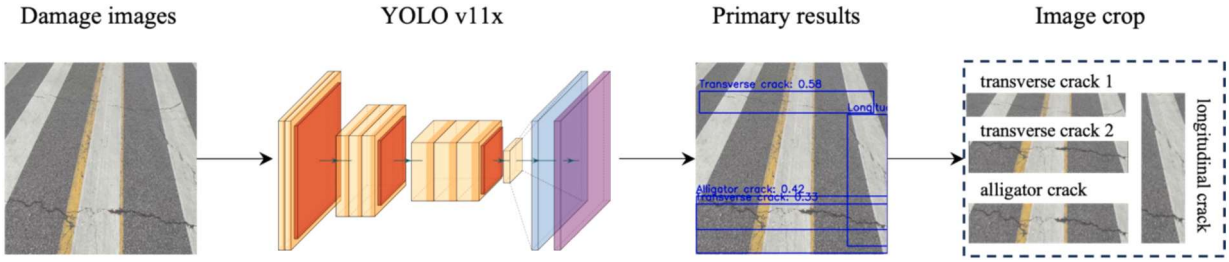


Fig. 2. Process for detecting defects.

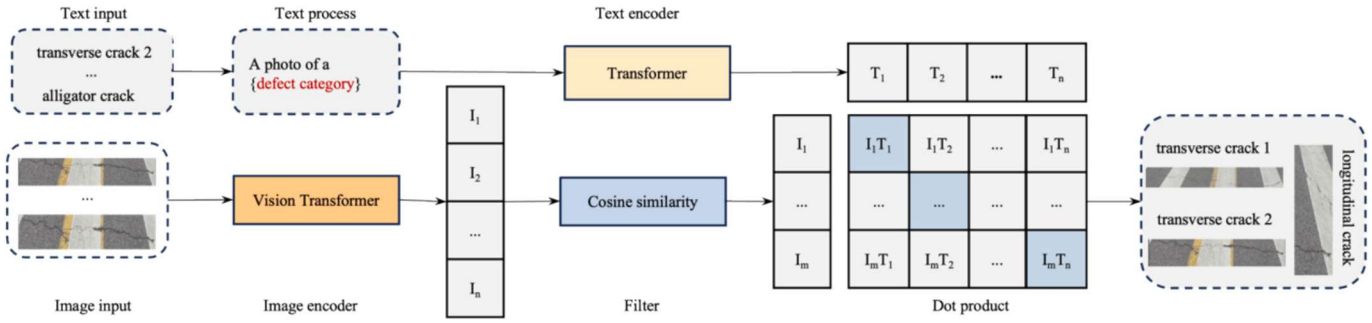


Fig. 3. CLIP-based repeat defects eliminate module.

information needed to be transformed first. For example, for an alligator, we transform it into "A photo of an alligator crack" before inputting it into the text encoder.

These text inputs are then processed by CLIP's Transformer-based text encoder [44]. This encoder transforms the input text into a normalized text feature vector $\boldsymbol{t}$ within a pre-trained multimodal embedding space.

Similarly, the defect image patches cropped from the YOLO detection boxes (resized to 224 × 224 pixels) are fed into CLIP's Vision Transformer (ViT) based image encoder [45]. This encoder converts the image patch into a corresponding normalized image feature vector $\boldsymbol{I}$, residing in the same embedding space as the text feature vectors.

As illustrated in Fig. 3, the detection results output by YOLO were first encoded using the Vision Transformer. For detection results, we obtained the encoded representations $[I_1, I_2, \ldots, I_n]$. Subsequently, the cosine similarity was calculated between each pair of image vectors. If the cosine similarity exceeded 0.95, the two detection results were considered identical. Through this process, the defects sequence was refined by filtering out duplicate detection objects, resulting in $[I_1, I_2, \ldots, I_m]$, where $m \leq n$.

3.2.3. Prompt generation

As shown in Fig. 3, the categories that text encoder needed to process include transverse cracks, longitudinal cracks, alligator cracks, and potholes. According to Chapter 3.2.2, we can get all the text features $t' = [t_1, t_2, t_3, t_4]$. The cosine similarity between the filtered arbitrary defects image features I_i ($I_i \in [I_1, I_2, \ldots, I_m]$) and the text features of the category can be used to measure the similarity between the image and the text. Finally I_i was processed using Softmax function to get the probability of I_i belonging to each category. The category with the highest probability was used by us as the final output. The process can be seen from (2)–(3).

$$F = \text{Similarity}(I_i, t') = I_i \cdot [t_1, t_2, t_3, t_4] = [f_1, f_2, f_3, f_4] \tag{2}$$

$$\text{Result} = \text{Softmax}(F) \tag{3}$$

This structured prompt serves to focus the VLM's attention and provide essential context (defect type, location) that goes beyond the raw image pixels, enabling it to generate more targeted and informative descriptions. This structured approach also allows for controllable and scalable prompt generation. The overall process of prompt generation can be seen in Fig. 4. The generated prompts primarily consist of class prompts, location prompts, and user prompts. For class prompts, the CLIP model eliminated duplicate detection results and identified the defect category. Regarding location information, the SORT object detection model YOLO v11x accurately determined the precise location of the defect. Users are only required to input simple prompts, such as "Describe this image." This design enables non-experts to utilize the model effectively for detecting pavement damage.

It is important to emphasize that the VLM receives the full original image as visual input concurrently while processing these prompts. The purpose of including precise coordinate information from YOLO V11X within the text prompt is to serve as a textual attention guidance mechanism. These coordinates act as explicit textual cues that can help guide the VLM's attention to focus on specific regions within the image identified as important by the detector. Furthermore, this approach helps ground the VLM-generated descriptions to specific detected

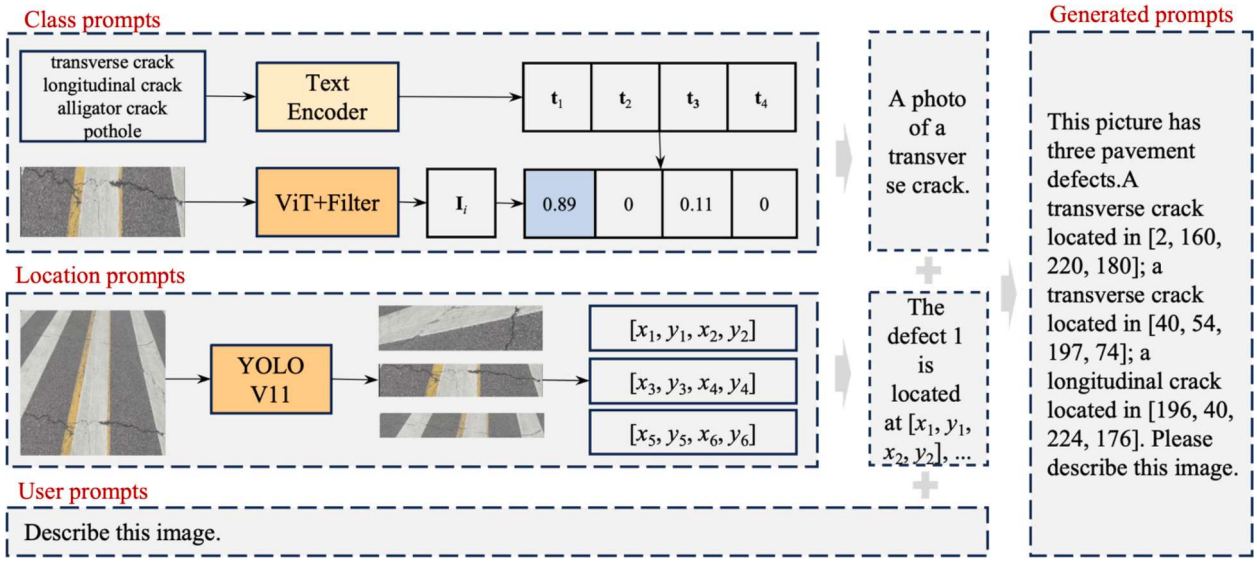


Fig. 4. Prompt generation.

instances, which is especially useful when multiple similar defects exist. This method effectively transfers the detector's high-precision localization strength to the VLM, enabling it to produce more specific and contextually aware descriptions about each identified damage.

The prompt generation method proposed in this study, which integrates visual information, demonstrates robust output controllability and scalability. Such a system can generate prompts using predefined templates, producing outputs that not only include the necessary technical details but also presented in natural language. In addition, the system can generate detailed prompts for technicians or summary prompts for managers. In terms of scalability, the model was portable and can be used to add new types of defects without having to change the overall framework. Furthermore, this scalability extended not only to inspection objects but also to the flexibility of replacing YOLO v11x with more advanced object detectors, thereby continuously enhancing the system's performance.

3.2.4. *Few-shot learning*

While our framework aims for strong zero-shot performance via effective prompting, we incorporate few-shot learning to further enhance descriptive accuracy and tailor the VLM's output to the specific nuances of pavement damage assessment with minimal data overhead. Building upon the visually-informed prompts generated in the previous steps, we utilize in-context learning to enhance the model's ability to process and understand new samples. Consequently, we developed a few-shot learning dataset comprising four samples, each containing pavement damage images accompanied by expert-provided descriptions. These descriptions include detailed information about the cracks, such as their location, size (in pixels), type, and an assessment of potential safety risks (see Fig. 5).

It is important to note that these relevant examples are not retrieved dynamically from a larger database for each test sample. Instead, they were pre-selected and fixed by experts. This means that when conducting a k-shot learning experiment, all test samples utilized the exact same set of k pre-selected examples as context (where $k = 1$, 2, or 4). This 'careful selection' process was primarily based on the following criteria: 1) Diversity of damage types: ensuring coverage of major types like transverse, longitudinal, and composite cracks; 2) Completeness of information content: each example's description included location, size, safety impact, and repair recommendation; 3) Representativeness of scenarios: choosing typical damage situations; 4) Image and text quality: ensuring image clarity and description accuracy; and 5) Progressive complexity: gradually introducing different scenarios from 1-shot to 4-shot. Based on these criteria, experts selected a small (total of 4, shown in Fig. 5) yet highly informative set of samples, aimed at facilitating effective in-context learning with minimal data. This approach allows the model to learn from a small but highly informative dataset, facilitating effective in-context learning with minimal data.

In the specific implementation of few-shot learning, we adopt a progressive learning model from 1-shot to 4-shot. Each shot contained a complete image-text pair, where the description text was provided by a professional and covered the key feature information of the damage. The samples can be seen from Fig. 5. For example, the description of the 1-shot sample delineates the location of the horizontal crack (situated in the lower left quadrant of the image) and its dimensions (approximately 200 pixels in length). The 2-shot sample then demonstrates the characteristics of the horizontal crack, while subsequent samples encompass cases of longitudinal and composite cracks. The progressive design of the sample ensures that the model is able to learn the feature representation of different types of defects.

To ensure the model's learning effectiveness, four essential elements are incorporated into the description annotation: the specific location of

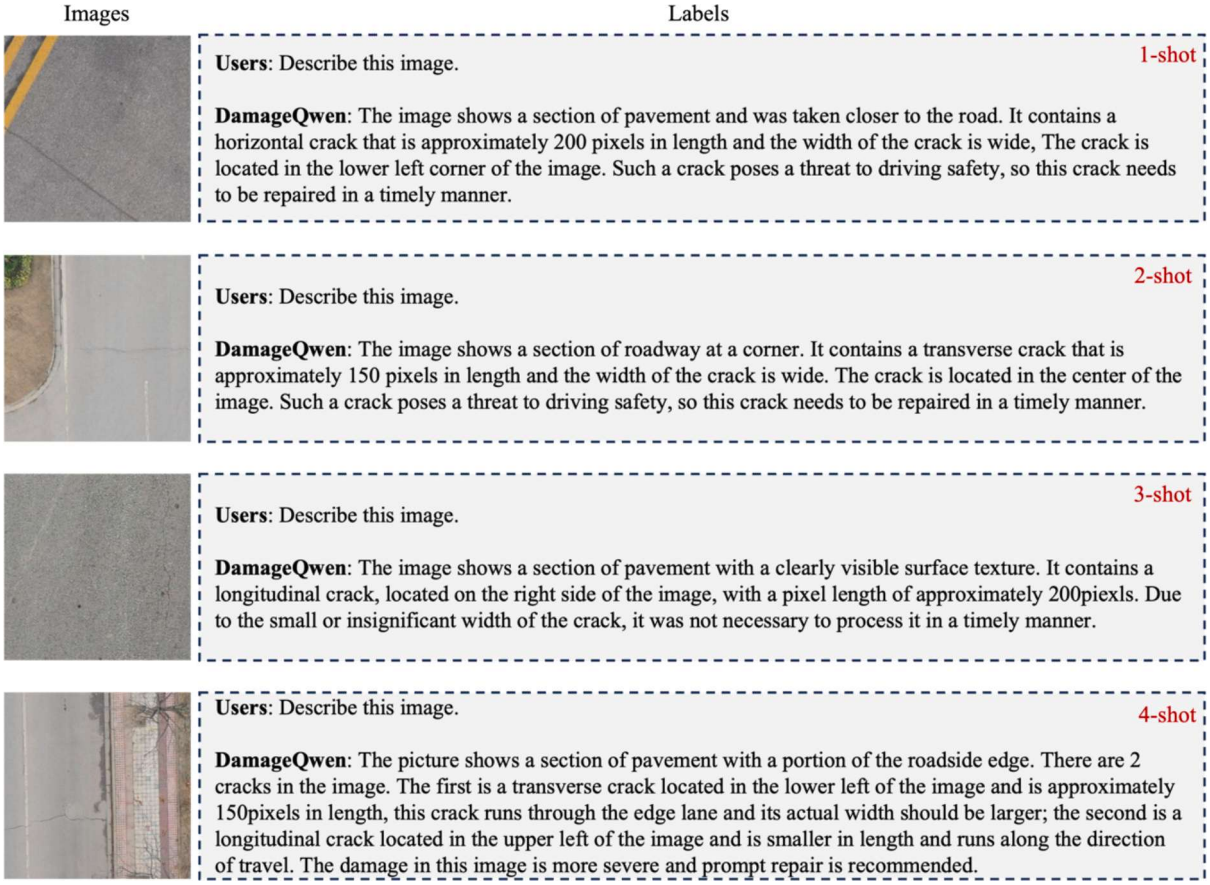


Fig. 5. Few-shot learning dataset.

the defect, a pixel-level measurement of its dimensions, an assessment of its potential impact on driving safety, and a corresponding repair recommendation. This structured annotation framework enables the model to develop a comprehensive understanding of pavement defects. By employing this well-designed few-shot learning paradigm, we aim for the VLMs to rapidly grasp the essential components of pavement damage detection and deliver effective recognition and assessment capabilities for real-world applications.

In 1-shot learning, the first sample was used to train the VLMs; in 2-shot learning, the first two samples were utilized; and in 4-shot learning, all four samples were employed to train the model. The model parameters remain frozen, and description accuracy is only enhanced by in-context examples.

4. Shadowed crack dataset

This section describes the datasets used for training and evaluating the DamageQwen model, specifically detailing the object detection dataset for YOLO v11x and the Vision-Language Model dataset used for both in-context learning and evaluation.

4.1. Object detection dataset

In our proposed DamageQwen model, we trained YOLO v11x to enhance pavement damage detection. To minimize the effort required for data labelling, we utilized the publicly available and extensive Road Damage Dataset (RDD2022) [26]. This dataset includes pavement damage images collected from multiple countries, featuring various types of damage such as transverse cracks, longitudinal cracks, alligator cracks, and potholes. The significant variability among images from different countries makes this dataset particularly suitable for training YOLO v11x, enabling the development of robust and accurate pavement damage detection models. The samples from RDD2022 is presented in Fig. 6.

4.2. Vision language models dataset

In addition to the dataset used for VLMs in-context learning introduced in Section 3.2.4, we also constructed a test set for evaluating the VLMs. The test set was still labeled using expert annotation. We annotated a total of 285 samples, as illustrated in Fig. 7. The test samples are saved in JSON format, with each sample comprising two main components: the image path (image_path) and the corresponding dialogue content (conversation). Within the dialogue structure, we designed an image description request initiated by a user role (role: 'user') and provided a standardized expert description (description) to serve as an evaluation benchmark.

Using the test sample illustrated in Fig. 7 as an example, the expert description provides detailed information about multiple defect features in the pavement. Specifically, it identifies two longitudinal cracks, each approximately 100 pixels in length, located at the top-right corner of the image, and a transverse crack, approximately 80 pixels in length,


```
{
  "image_path": "test/00260.jpg",
  "conversation": [
    {
      "role": "user",
      "content": [
        {
          "type": "image",
          "image": "test/00260.jpg"
        },
        {
          "type": "text",
          "text": "Describe this image."
        }
              ]
    }
],
"description": "The image shows several pavement defects. There are two longitudinal cracks located at the top-right corner with a pixel length of approximately 100 pixels each. Additionally, there is one transverse crack near the center of the image with a pixel length of around 80 pixels. These defects should be repaired due to their size and potential impact on safety and aesthetics.\n There are no visible alligator cracks or potholes in this image. Therefore, based on the described defects, repairs are recommended for the longitudinal and transverse cracks."
}
```


Fig. 7. Test set sample.

situated at the center of the image. The description not only specifies the location and size of the defects but also explains the necessity of repair, considering their size and potential impact on safety and aesthetics. In addition, the description explicitly states that no other types of damage, such as alligator cracks or potholes, were observed in the image. This level of completeness ensures a comprehensive assessment of the model's recognition capabilities.

This structured test sample design not only enables the quantitative evaluation of model performance but also serves as a reliable reference standard for assessing the model's applicability in real-world scenarios. By comparing the consistency between the model outputs and the expert descriptions, we can comprehensively evaluate the model's performance across various aspects, including damage location identification, size estimation, and damage assessment.

It should be emphasized that the descriptive labels for the few-shot examples used for VLM in-context learning and for the test set described in this section were generated through manual expert annotation, independently of the bounding box annotations from the RDD2022 dataset used for YOLO training. The pixel estimations for crack length included in the descriptions were obtained by experts using image measurement tools to manually measure visually along the main path of the crack. These serve as approximate values intended to provide



Fig. 6. RDD2022 dataset.

size reference for the VLM and for evaluation purposes.

4.3. Evaluation metrics

To comprehensively evaluate the effectiveness of model generation, this paper employed several widely adopted evaluation metrics. First, we utilized the ROUGE (Recall-Oriented Understudy for Gisting Evaluation) family of metrics, including ROUGE-1, ROUGE-2, and ROUGE-L, to calculate F1 scores for single words, two-word combinations, and the longest common subsequence, respectively. The ROUGE metrics are employed to evaluate the fluency and accuracy of the generated text by comparing it with the reference text at different n-gram levels. Specifically, ROUGE-1 measures word-level matching, ROUGE-2 evaluates phrase coherence, and ROUGE-L captures overall structural similarity through the longest common subsequence.

In addition, we used the BLEU (Bilingual Evaluation Understudy) score as a supplementary evaluation metric, which was based on a weighted geometric mean of n-gram accuracy and introduced a brevity penalty factor to avoid too-short outputs, and which was initially used for machine translation evaluation and is now widely used in various text generation tasks. To further enhance the reliability of the evaluation, we also utilized the METEOR metric, which captures more subtle semantic similarities by considering stemming and synonyms on top of word-level matching.

Finally, considering that the above metrics may not adequately reflect the deep semantic information, we introduced BERTScore based on a pre-trained language model. BERTScore leverages BERT encodings to compute textual representations and evaluates the semantic similarity between generated and reference texts at the token level using cosine similarity. By incorporating BERTScore, the evaluation process gains a more refined assessment of contextually relevant semantic information, thereby offering a more comprehensive quality evaluation. The integration of these metrics ensures both the reliability and the comprehensiveness of the assessment results.

For object detection models, we still used mAP, mAP50–95, Precision and Recall as evaluation metrics.

4.4. Model training process

The settings of all the hardware, software and hyperparameters used in the training process of this study are shown in Table 1. For the object detection algorithms, experimental results shown that 100 epochs are sufficient to train the model to convergence.

5. Results and discussion

This section analyses the experimental results, including the performance of object detection models, evaluation of DamageQwen in various learning scenarios, impact of different detectors, ablation studies, and practical test outcomes.

Table 1
Software and hardware versions.

Content	Versions
CPU	Intel Xeon Processor (Skylake, IBRS)
GPU	NVIDIA Tesla V100-32GB
Operating system	Ubuntu 22.04
Python	3.10
PyTorch	2.1.2
RAM	72GB
Cuda	11.8
Learning rate	1e-4
Batch size (object detection)	16
Epochs (object detection)	100
Momentum (object detection)	0.937
Number worker (object detection)	4
Optimizer	Adam

5.1. SORT object detection model

The one-stage real-time object detection models, the YOLO series, were utilized in this study. Fig. 8 illustrates the performance of different YOLO model versions during training. It comprises four subgraphs: mAP50, mAP50–95, Precision, and Recall. From the overall trends observed in these subgraphs, all metrics demonstrate an upward trajectory as the number of training epochs increases, eventually stabilizing between 60 and 100 epochs. This indicates that all models have reached a converged state.

The results from the 100th training epoch are presented in Table 2. It can be observed that the performance of small-parameter models (n-series) is inferior to that of large-parameter models (x-series) within the same version. Among all models, the newly released YOLO v11x achieves the best performance across all four metrics, demonstrating that the YOLO v11 architecture significantly enhances performance. This makes YOLO v11x a state-of-the-art (SOTA) model in the field of object detection. In scenarios where computational resource constraints are not a concern, YOLO v11x is the optimal choice for object detection tasks. Consequently, YOLO v11x was employed as the pavement damage detection model in this study.

5.2. Results of zero-shot and few-shot learning

In this section, we provide a comprehensive evaluation of the DamageQwen model, which was developed based on the Qwen2-VL-7B-Instruct-GPTQ-Int8 architecture [46,47], and compare it to several state-of-the-art VLMs, including the original Qwen2-VL-7B-Instruct-GPTQ-Int8, Qwen2-VL-2B-Instruct and Llama-3.2-11B-Vision [48]. The evaluation process was performed in zero-shot and few-shot, respectively. In addition, we also used two different prompt strategies.

Under the first "*describe this image*" prompt strategy, DamageQwen consistently outperforms the comparison model across nearly all evaluation metrics, the results can be seen from Table 3. Notably, in the 4-shot learning, DamageQwen demonstrates substantial performance enhancements, achieving ROUGE-1 at 0.4769, ROUGE-2 at 0.1562, and ROUGE-L at 0.3692. These results represent a significant improvement over the baseline model.

DamageQwen demonstrates greater performance improvements when employing the second detailed prompt strategy ("*How many pavement defects are present in the image?(Including longitudinal cracks, transverse cracks, alligator cracks, and potholes). For each defect, please describe: Location (right, left, top, bottom, top-right, bottom-right, top-left, bottom-left, or center of the image), Geometric characteristics (the pixel length of longitudinal cracks and transverse cracks, the pixel area of alligator cracks and potholes). Should these defects be repaired?*"). The results can be seen from Table 4. The model achieves its highest performance under the 4-shot learning, with ROUGE-1 at 0.4828, ROUGE-2 at 0.2162, and ROUGE-L at 0.3701. This structured prompt approach encourages the model to output more precise and detailed descriptions, as reflected in the higher BLEU score (0.1494) and Meteor score (0.3494). Additionally, the model achieves an excellent Bert-score of 0.9099.

As can be seen in Table 4, DamageQwen using 4-shot learning achieves the best performance. The high ROUGE-1 (0.4828) and ROUGE-2 (0.2162) scores indicate that the model effectively captures word-level and two-word phrase-level matches, demonstrating a strong lexical consistency between the generated descriptions and the reference text. Similarly, the strong performance of ROUGE-L (0.3701) suggests that the generated descriptions align well with the reference text in terms of the longest common subsequence, highlighting the model's ability to maintain coherent sentence structures. However, the relatively low BLEU score (0.1494) arises primarily from discrepancies in formatting; while the reference text is often presented as paragraphs, the output from VLMs is sometimes structured line-by-line. Although the model generates accurate content, differences in expression contribute to the lower BLEU score. The enhancement in the Meteor score reflects the

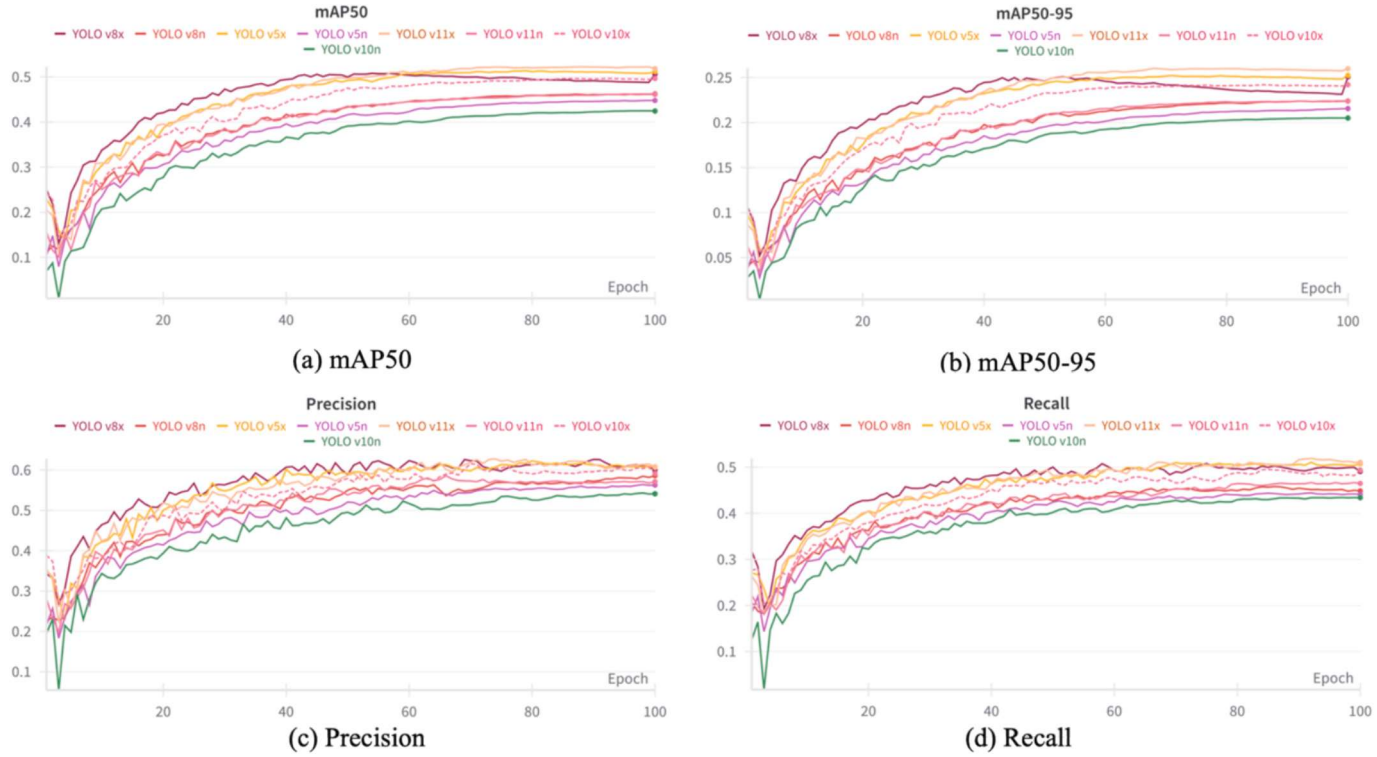


Fig. 8. Results of different object detection models.

Table 2
Results of different object detection models.

Model	mAP50	mAP50–95	Precision	Recall
YOLO v5n	0.4478	0.2155	0.5627	0.4411
YOLO v5x	0.5118	0.2524	0.6067	0.5076
YOLO v8n	0.4623	0.2238	0.5861	0.4486
YOLO v8x	0.5059	0.2509	0.5999	0.4907
YOLO v10n	0.4243	0.2050	0.5412	0.4340
YOLO v10x	0.4965	0.2421	0.5912	0.4937
YOLO v11n	0.4611	0.2241	0.5711	0.4648
YOLO v11x	**0.5183**	**0.2600**	**0.6078**	**0.5105**

model's proficiency in synonym recognition and semantic matching. This suggests that the generated descriptions are more closer to human expressions at the semantic level. Furthermore, the high Bert-score (0.9099) indicates that the model-generated descriptions exhibit a high degree of similarity to the reference text in terms of contextual semantic representation.

From the comparison between Tables 3 and 4, we can realize that few-shot learning is also effective in the field of pavement damage descriptions. The performance of all models show a continuous improvement as the number of shots increases. Moreover, providing more detailed prompts enables the model to produce more accurate and comprehensive descriptions of the pavement damage. The visually enhanced prompting strategy, based on YOLO and CLIP, proposed in this study significantly enhances the model's performance. First, the object detection model YOLO v11x delivers precise damage category and location information, offering a reliable spatial reference for prompt generation. Second, the multimodal feature extraction capability of the CLIP model filters out duplicate defects detected by YOLO v11x, thereby improving the accuracy of category and location in the prompt words. Finally, by integrating this structured information with the user prompt, the model is able to generate more realistic and descriptive content.

These results confirm the effectiveness of the proposed visual

Table 3
Performance comparison of different models (prompt 1).

shots	Model	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	Meteor	Bert-score
0-shot	Qwen2-VL-2B-Instruct	0.3099	0.076	0.1987	0.027	0.2056	0.8542
	Qwen2-VL-7B-Instruct	0.3055	0.0687	0.1849	0.0228	**0.2265**	0.8502
	Llama-3.2-11B-Vision	0.2997	0.0692	0.2011	0.0225	0.2017	0.8553
	DamageQwen	**0.3354**	**0.1031**	**0.2172**	**0.0369**	0.2173	**0.8626**
1-shot	Qwen2-VL-2B-Instruct	0.3695	0.1164	0.2477	0.0438	0.1941	**0.8767**
	Qwen2-VL-7B-Instruct	0.3368	0.1027	0.2255	0.0412	0.1919	0.8679
	Llama-3.2-11B-Vision	0.3778	0.0787	0.2222	0.0431	0.2003	0.8746
	DamageQwen	**0.3946**	**0.1176**	**0.26**	**0.0513**	**0.2232**	0.8719
2-shot	Qwen2-VL-2B-Instruct	0.3897	0.0984	0.2398	0.039	0.2143	0.8861
	Qwen2-VL-7B-Instruct	0.3931	0.1062	0.2421	0.0456	0.2174	0.8857
	Llama-3.2-11B-Vision	0.3958	0.1158	0.2708	0.0563	0.2314	0.871
	DamageQwen	**0.458**	**0.155**	**0.3194**	**0.1014**	**0.3085**	**0.9075**
4-shot	Qwen2-VL-2B-Instruct	0.3999	0.1121	0.2588	0.044	0.2149	0.8873
	Qwen2-VL-7B-Instruct	0.4052	0.1248	0.2665	0.0527	0.22	0.8909
	Llama-3.2-11B-Vision	0.3842	0.1194	0.3054	0.0463	0.2035	0.8799
	DamageQwen	**0.4769**	**0.1562**	**0.3692**	**0.1089**	**0.3972**	**0.9071**

Table 4
Performance comparison of different models (prompt 2).

shots	Model	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	Meteor	Bert-score
0-shot	Qwen2-VL-2B-Instruct	0.4044	**0.1625**	0.2703	0.0582	0.2625	0.8695
	Qwen2-VL-7B-Instruct	0.4178	0.1559	0.2708	0.0435	0.2595	0.867
	Llama-3.2-11B-Vision	0.4103	0.1034	0.2137	0.036	0.2247	**0.8762**
	DamageQwen	**0.4554**	0.1578	**0.3098**	**0.0726**	**0.272**	0.8671
1-shot	Qwen2-VL-2B-Instruct	0.3806	0.1243	0.2734	0.0467	0.2707	0.8712
	Qwen2-VL-7B-Instruct	0.4269	0.1413	0.275	0.0341	0.2499	0.8737
	Llama-3.2-11B-Vision	0.4229	0.1156	0.2629	0.0671	0.2634	0.8766
	DamageQwen	**0.4676**	**0.1671**	**0.3482**	**0.1234**	**0.2991**	**0.881**
2-shot	Qwen2-VL-2B-Instruct	0.4179	0.1564	0.2733	0.0322	0.2943	0.8689
	Qwen2-VL-7B-Instruct	0.4554	0.1689	0.2942	0.0752	0.2757	0.8868
	Llama-3.2-11B-Vision	0.4211	0.1527	0.2857	0.0967	0.2753	0.8775
	DamageQwen	**0.4662**	**0.191**	**0.3506**	**0.1332**	**0.3447**	**0.9093**
4-shot	Qwen2-VL-2B-Instruct	0.4266	0.1642	0.272	0.0837	0.2846	0.8854
	Qwen2-VL-7B-Instruct	0.4378	0.1852	0.3113	0.087	0.2941	0.8914
	Llama-3.2-11B-Vision	0.4372	0.1829	0.2732	0.0961	0.2873	0.8872
	DamageQwen	**0.4828**	**0.2162**	**0.3701**	**0.1494**	**0.3494**	**0.9099**

enhancement framework and demonstrate that integrating accurate damage detection results with VLMs significantly enhances the accuracy and completeness of road damage descriptions. This approach not only improves the performance of the model, but also provides a new paradigm for the development of automated pavement inspection systems.

5.3. Results of different object detection model

To verify the scalability and generality of our proposed framework, we conducted systematic comparative experiments on different versions of YOLO object detectors. The results of the experiments are shown in Table 5, where we selected several mainstream versions from YOLOv5x to YOLOv11x for evaluation. It is worth noting that the data in Table 5 was obtained in the zero-shot with Prompt 1 configuration.

The results show that our proposed framework adapts well to different versions of the YOLO detector with relatively stable performance differences. YOLO v11x achieves best performance across several metrics, with ROUGE-1 at 0.3354, ROUGE-2 at 0.1031, and ROUGE-L at 0.2172. These results suggest that the latest version of the detector provides more accurate defects localization and feature information. Combined with the results in Table 2, it is evident that the performance differences primarily arise from the accuracy of the object detector. This adaptability allows the system to flexibly integrate the SORT object detection technologies, thereby continuously enhancing the accuracy and completeness of pavement damage descriptions. Furthermore, this scalable design supports the long-term evolution of the framework in practical applications. Although variations in accuracy exist among different object detection models, these differences have a minimal impact on the performance of DamageQwen's damage descriptions. Overall, a visual enhancement framework based on object detectors significantly outperforms a single VLM.

Table 5
Performance comparison of YOLO models.

Model	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	Meteor	Bert-score
YOLO v5x	0.3296	0.0995	0.2071	0.0341	0.2142	0.8621
YOLO v8x	0.3325	0.1011	0.2089	0.0344	0.213	**0.8632**
YOLO v10x	0.3287	0.0969	0.2063	0.0337	0.2114	0.8605
YOLO v11x	**0.3354**	**0.1031**	**0.2172**	**0.0369**	**0.2173**	0.8626

5.4. Results of ablation studies

To obtain a deeper understanding of the contribution of each component within the framework, we conducted detailed ablation experiments. By sequentially incorporating the YOLO and the CLIP model, we systematically evaluated the impact of each component on the overall performance. The results of these evaluations are presented in Table 6. The baseline model (Baseline) refers to using only the core Qwen2-VL-7B-Instruct-GPTQ-Int8 VLM with the simple "Describe this image." prompt and the raw image input, without any enhancement from YOLO or CLIP.

The baseline model exhibited relatively low performance metrics, reflecting the limitations of relying on a VLM alone to accurately describe pavement damage. The introduction of the YOLO v11x resulted in significant improvements in all metrics: ROUGE-1 improved to 0.3312, ROUGE-2 reached 0.0973, and the BLEU score increased to 0.0339. This is because the precise spatial localization information provided by the YOLO v11x, which enables the model to more accurately locate and describe the location of the defect. This ablation analysis aimed to quantify the respective contributions of the YOLO-based prompt enhancement and the CLIP-based result refinement to the overall description performance by progressively introducing these key components. The results clearly indicate that both components provide a positive and measurable impact on improving the quality of the final output.

Furthermore, the introduction of the CLIP brings additional performance improvements. Although the improvement over YOLO is small, the main contribution of CLIP is to utilize its multimodal feature representation to effectively remove the duplicate detection boxes that could not be completely eliminated by non-maximal suppression (NMS) in the YOLO model. This optimization ensures that the final generated prompts are more concise and accurate, avoiding repeated descriptions

Table 6
Results of ablation studies.

Model	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	Meteor	Bert-score
Baseline	0.3055	0.0687	0.1849	0.0228	**0.2265**	0.8502
Baseline + YOLO v11x	0.3312	0.0973	0.2154	0.0339	0.2151	0.8613
Baseline + YOLO v11x + CLIP	**0.3354**	**0.1031**	**0.2172**	**0.0369**	0.2173	**0.8626**

of the same damage. The effectiveness of this enhancement is confirmed by the continuous improvement in the Bert-score, which ultimately reaches 0.8626, indicating that the generated descriptions are more coherent and accurate at the semantic level.

The results of the ablation studies confirm the robustness and necessity of the integrated design of the DamageQwen framework. Each component contributes a unique function, creating a synergistic effect: the baseline model establishes fundamental visual-language comprehension, the YOLO model provides precise spatial localization information, and the CLIP feature extractor enhances detection results through its multimodal feature representation capability.

In total, these components synergistically form an efficient and effective pavement damage description framework, and the results of this ablation study provide strong justification for their inclusion in the framework's architecture.

Compared to traditional object detection method, we emphasize that the two approaches aim to solve tasks at different levels and of different types. Traditional object detection focuses primarily on localization and classification, outputting bounding boxes and labels, suitable for scenarios requiring rapid identification of damage locations. However, it cannot provide detailed descriptions, contextual information, severity assessment, or interaction via natural language. DamageQwen aims to move beyond detection towards description and preliminary assessment. By integrating the localization precision of YOLO, the multimodal filtering capability of CLIP, and the language generation and understanding power of a VLM, our framework can output rich natural language reports, answer user questions, and make the damage status interpretable even for non-experts. While direct quantitative metric comparison (e.g., mAP vs. ROUGE) is not feasible due to the different tasks, the conceptual differences and qualitative results (e.g., Figs. 9 and 10) demonstrate significant added value provided by DamageQwen, especially in application scenarios requiring detailed reporting, human-computer interaction, and broader user accessibility. Therefore, we argue that the introduced architectural complexity is justified and necessary to achieve these advanced capabilities.

5.5. Results of the practical test

In order to further verify the performance of DamageQwen, we conducted practical tests, and the results are shown in Figs. 9 and 10. These tests were done under zero-shot learning. As shown in Fig. 9, the three models (Llama-3.2-11B-Vision, Qwen2-VL-7B-Instruct, and DamageQwen) demonstrate different recognition capabilities and levels of detailed description. The Llama model demonstrated limited capability, identifying only the basic features of the image, such as grayscale and granular texture, but failing to detect the critical feature of cracks. In contrast, the Qwen2-VL-7B-Instruct model exhibited superior performance by not only identifying the presence of horizontal cracks but also describing the specific material characteristics of the surface, suggesting it could be stone or concrete. The DamageQwen model showcased the most advanced analytical ability, accurately identifying two cracks (horizontal and longitudinal), providing detailed descriptions of their exact locations and approximate dimensions (around 100 pixels in length, although this result is wrong), and offering professional maintenance recommendations.

The performance of all three VLMs in the pavement crack identification task improved significantly when provided with detailed prompt. The Llama-3.2-11B-Vision model successfully identified major longitudinal crack, provided basic spatial localization information, and offered repair recommendations. However, its descriptions were relatively simple and limited to identifying a single crack. In contrast, the Qwen2-VL-7B-Instruct model exhibited a more cautious and professional analytical approach. It not only identified the location and type of cracks but also highlighted the impact of image resolution limitations on crack measurements. Nevertheless, neither of these models was capable of identifying all the cracks. By leveraging the proposed visual enhancement framework, DamageQwen not only accurately identified two cracks but also provided detailed descriptions of their geometrical characteristics, such as width variations and curvature degrees. Additionally, it offered professional recommendations to prevent further deterioration.

These results confirm the significant advantages of specialized VLMs, particularly those based on visual augmentation frameworks, over generic VLMs. DamageQwen's performance is especially noteworthy, highlighting the potential of specialized domain-specific models to deliver detailed, accurate, and practically applicable analyses.

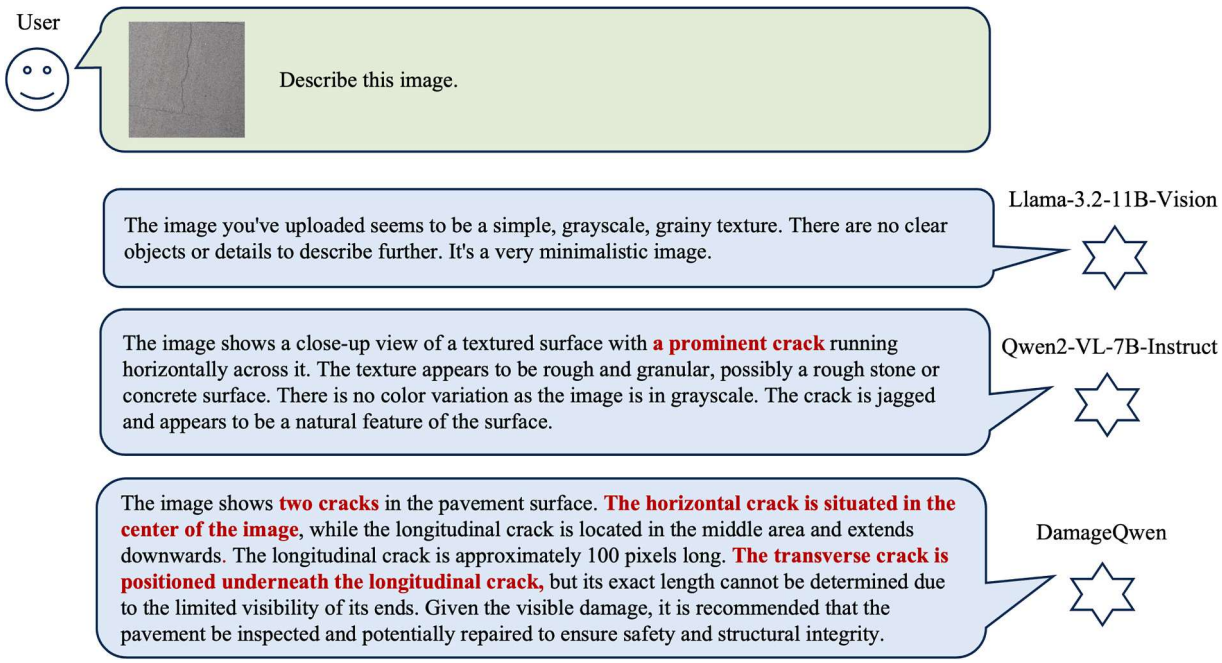


Fig. 9. Test results for simple prompt.

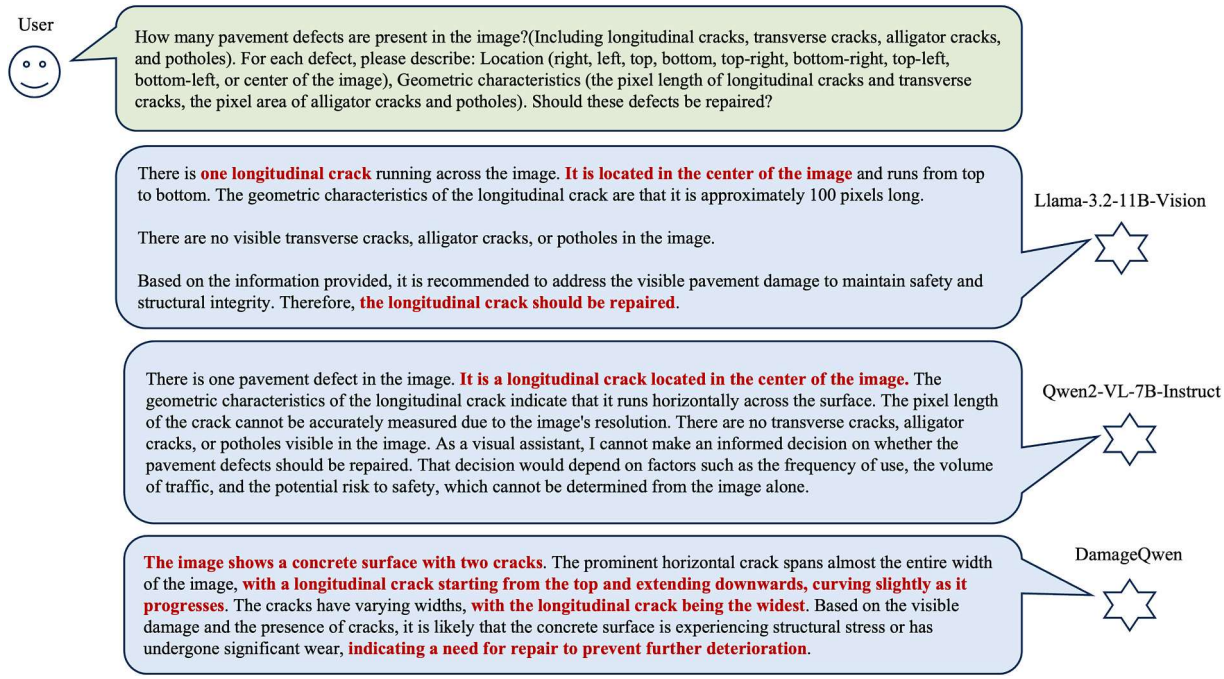


Fig. 10. Test results for detailed prompt (Red font indicates key descriptive phrases and assessments from the model's output). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

6. Conclusions

This paper presented DamageQwen, an framework that combined the strengths of traditional object detection models with VLMs for pavement damage detection. The main conclusions can be summarized as follows:

1. The integration of YOLO v11x and VLMs creates a synergistic effect that overcomes the limitations of each individual approach. While YOLO v11x provides precise damage detection, the VLMs component enhances the system's ability to understand complex contextual information and damage characteristics. DamageQwen outperforms generic VLMs (Llama and Qwen) for pavement damage ability in either zero-shot or few-shot learning.
2. The CLIP-based repeat defects elimination module, as a key part of the integrated framework, effectively addresses the challenge of redundant detection boxes, particularly in cases where different categories are assigned to the same damage feature. This improvement enhances the overall accuracy and reliability of the framework.
3. By enabling natural language interaction through visual question answering, DamageQwen makes pavement damage assessment more accessible to non-expert users while maintaining professional-level accuracy. This feature represents a significant step towards democratizing infrastructure inspection processes.

In conclusion, DamageQwen is more than just a damage detection tool; it represents a shift towards a more comprehensive, intelligent, and accessible paradigm for infrastructure assessment, combining precise visual recognition with deep language understanding to provide more valuable information support for pavement maintenance decisions.

While DamageQwen shows promise, limitations exist. First, current severity assessment relies on qualitative descriptions and approximate VLM-based measurements (e.g., pixel length), lacking a quantitative standard. Second, the framework's generalizability to unseen damage types or significantly different environmental conditions needs further validation. Furthermore, performance depends on the accuracy of upstream components (e.g., YOLO detection, CLIP filtering). Finally, the multi-model integration may incur significant computational costs. Future work should focus on developing quantitative assessment metrics, enhancing model robustness and generalizability, and optimizing computational efficiency.

CRediT authorship contribution statement

Yingchao Zhang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Cheng Liu:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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